ME594: 3D Robotics, HW 2
Homework due on 01-26-2025, Topics: Euler Anlges.

1. **Homogenous Transformations:** The frame $x_0y_0z_0$ is at the origin. Then, it undergoes homogenous transformation to $x_1y_1z_1$ and finally $x_2y_2z_2$.

   (a) Compute the homogenous transformations H_1^0, H_2^0, H_2^1.

   (b) Confirm the identity numerically: $H_2^0 = H_1^0 H_2^1$.

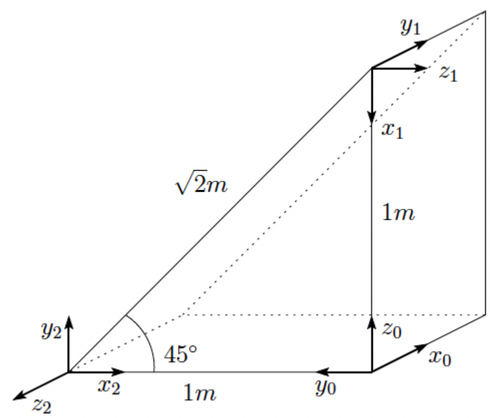

Figure 1: Homogenous Transformations

   - **Gradescope:** Submit your typed/handwritten work. Feel free to use Python if needed.
   - **Email:** None

2. **Euler angles:** Consider the Euler angles ϕ, θ, ψ denoting rotation about the x-, y-, and z-axis. The net rotation $z - y - x$ or $3 - 2 - 1$ is given by

$$\begin{aligned}\mathbf{R} &= \mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi) \\ &= \begin{bmatrix} 0.4330 & -0.4356 & 0.7891 \\ 0.7500 & 0.6597 & -0.0474 \\ -0.5000 & 0.6124 & 0.6124 \end{bmatrix}.\end{aligned}$$

   Find the numerical values of ϕ, θ, ψ.
   HINT: Expand R symbolically and then compare.

   - **Gradescope:** Submit your typed/handwritten work. Feel free to use Python if needed.
   - **Email:** None

3. **Pendulum waves in MuJoCo**
   Your goal is to use MuJoCo to recreate the wave pattern observed in this YouTube video: https://youtu.be/yVkdfJ9PkRQ. The video shows 15 simple pendulums that produce a wave pattern when launched. The lengths of the pendulum are chosen in a special way. The 1st pendulum executes 51 oscillations in 1 min; the 2nd pendulum executes 52 oscillations in 1

min, ..., and the 15th pendulum executes 65 oscillations. The lengths of the pendulums can be obtained by this formula,

$$\ell_i = \left(\frac{60}{50+i}\right)^2 \frac{g}{4\pi^2} \quad \text{where } i = \mathit{1}, \mathit{2}, ..., \mathit{15}$$

HINTS:

(a) You can specify two geometries (<geom> cylinder and sphere) within <body> tag to create the desired visual effect. Then using <inertial> tag you can specify the mass and position of the center of mass to be at the location of the sphere.

(b) This video shows how to model a simple pendulum. Specifically note how body frame is defined and then used to specify the pin joint and center of mass: https://youtu.be/S-374Yy6ObE.

(c) Once you create a single pendulum you can then copy paste it and modify the lengths using the formula given above.

(d) Here is my attempt on this problem: https://youtube.com/shorts/UG93vWeTd1g.

Submission:

- **Gradescope:** None.
- **Email:** Submit the XML file to pranav@uic.edu. Please add "HW2 Pendulum Waves" to your subject line. If submitting as a group, please copy your group member.